

Task 3: Automation System Design

Technical Design Report for an Automated Sensor-Driven Conveyor Sortation System

Project Domain: Mechatronics & Factory Automation | **Design Focus:** Logic Flow & System Specifications |
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1. System Overview & Functional Objectives

In modern manufacturing facilities, logistics sortation stands out as a critical production bottleneck. Manual handling lines slow down distribution, increase product tracking mistakes, and present workplace safety risks when moving heavy materials. To resolve these operational issues, this technical report documents an automated, sensor-driven conveyor sortation system built to achieve high-throughput classification without human intervention.

The primary architectural goals of this automation setup are to monitor components in motion, identify their geometric height profiles, check for structural metallic signatures, and run an electromechanical sorting gate. By detecting and pulling non-conforming or anomalous products off the line into an inspection cell, the system acts as a reliable quality control filter, letting only valid products pass uninterrupted downstream.

2. Hardware Core Architecture

The hardware architecture relies on a structured sequence of discrete mechatronic elements. Instead of complex software code, system stability depends on choosing the correct sensors and configuring precise electromechanical timings relative to the belt speed. The physical setup consists of a structural frame, a variable-speed motor drive, non-contact feedback sensors, and a gate actuator.

Component Classification	Hardware Specification	Primary System Functionality
Central Logic Unit	Industrial Embedded Microcontroller / Micro-PLC	Processes raw sensor inputs, coordinates system logic timing windows, and triggers output states.
Conveyor Transport Drive	12V DC High-Torque Gear Motor	Maintains a constant physical belt velocity to ensure reliable travel times between tracking nodes.
Primary Presence & Height Sensor	Ultrasonic Distance Sensor Node (HC-SR04)	Mounted vertically above the intake line to continually check object clearance profiles.
Secondary Composition Sensor	Inductive Proximity Sensor Node (LJ12A3)	Generates a localized electromagnetic fields to detect metal signatures on the fly.
Mechanical Actuator	High-Torque Metal Gear Servo Diverter Arm	Sweeps across the sorting line when triggered to physically steer non-conforming items away.

Table 1: Automation System Hardware Blueprint.

3. Process Logic and System Flowchart

The system runs on a continuous feedback sequence. At rest, the controller polls the vertical ultrasonic distance sensor to establish a reference height. The moment an object blocks this path, the measured distance drops below the baseline, marking the start of a sorting sequence. As the object passes the intake zone, the controller samples the height profile and cross-checks the inductive proximity sensor to verify material properties.

If the item matches all quality guidelines (e.g., standard sizing and non-metallic composition), the gate remains neutral, letting the product continue down the main line. If any check fails, the controller calculates the travel time to the gate based on conveyor speed and swings the diverter arm out at the exact millisecond the item arrives. The detailed tracking logic and decision paths are visualized below:

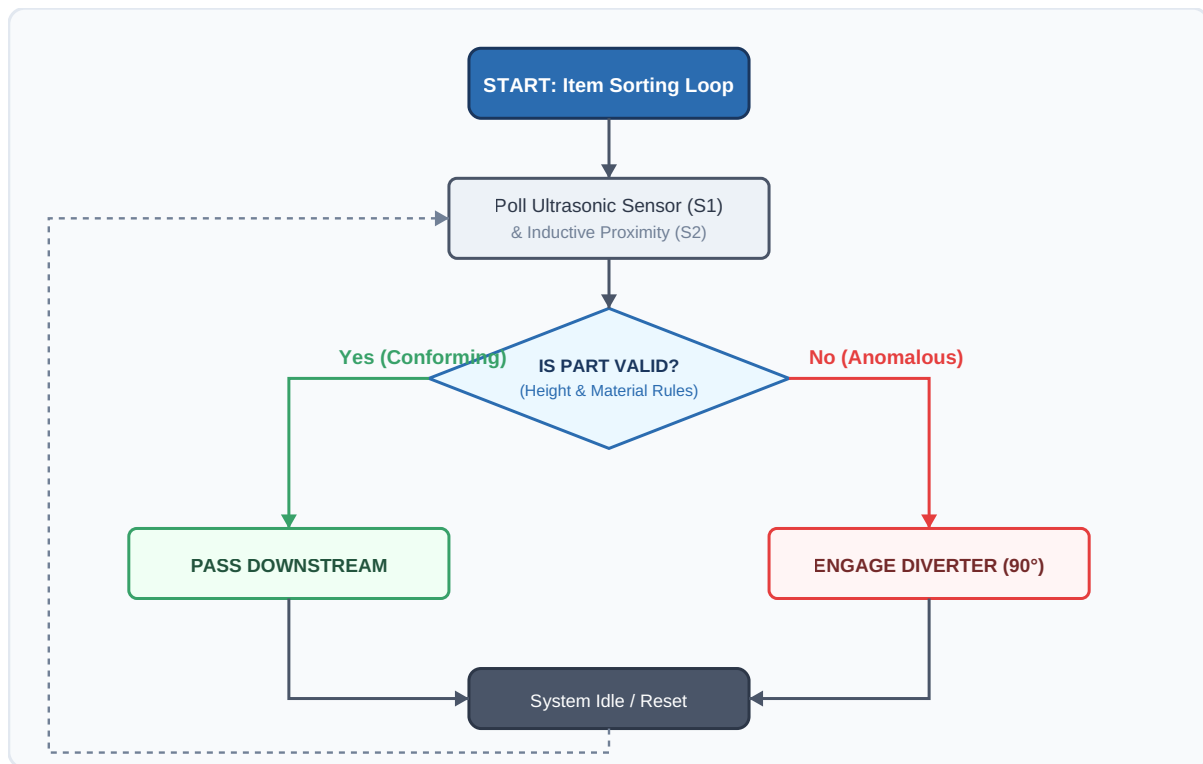


Figure 1: Programmatic Logic State Machine and Sensor Diverting Flowchart.

4. Detailed Explanatory Analysis

4.1 Data Input and Filtering Stage

The system accuracy starts at the edge sensor level. The primary controller communicates with the vertical ultrasonic transducer using high-frequency acoustics. The sensor emits an ultrasound pulse and measures the echo time of flight to find surface distance. The controller filters out brief noise or vibrations to avoid false triggers, confirming an item is present only when distance drops consistently over multiple tracking checks.

Simultaneously, the inductive sensor uses an internal oscillator circuit to track changing eddy currents. When a metallic item passes through this electromagnetic zone, it draws energy from the field, reducing the oscillation amplitude. The sensor converts this drop into a clean binary logic signal (High for non-metal, Low for metal), feeding it straight to the controller's digital registry.

4.2 Timing and Actuation Synchronization

Because sorting happens downstream from the initial scanning zone, the system synchronizes the actuator using physical travel equations rather than instant commands. The controller uses a mechanical equation to compute the arrival window:

$$\text{Time Delay} = \text{Fixed Tracking Distance} / \text{Conveyor Belt Speed}$$

Once an object triggers a fail state, an internal hardware timer starts a countdown based on this calculated value. When the timer hits zero, a Pulse Width Modulation (PWM) signal commands the high-

torque servo motor to swing exactly 90 degrees. This shifts the diverter arm into the path of the oncoming item, sliding it safely into the inspection bin. A brief reset window follows, clearing the track and returning the arm to its baseline position (0°) before the next component arrives.

5. Operational Calibration & Maintenance Profile

To maintain high accuracy over long production shifts, the automation cell requires regular mechanical maintenance. Dust can accumulate on the ultrasonic sensor grids, blocking acoustic paths and distorting height readings. For this reason, sensors must be cleaned and re-calibrated weekly against standard physical block dimensions.

Additionally, keeping the conveyor belt at a stable physical tension prevents belt slippage, which can throw off travel timing and cause the sorting arm to trigger too early or too late. By running these simple maintenance checks, the system delivers high sortation reliability, isolates production anomalies early, and provides stable, automated quality tracking data.